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The Quality Ceiling Is the Task, Not the Model Size

A Reproducible CPU/GPU Benchmark of Search-Relevance Judges

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Metric: Quadratic Weighted Kappa (QWK), 95% CI

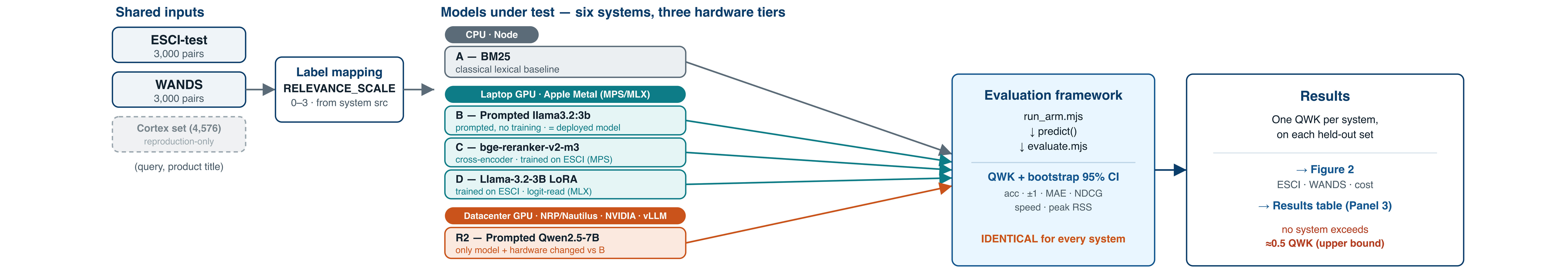

Figure 1. One evaluation framework, six systems, three hardware tiers. Shared inputs are label-mapped, then each **model under test** (CPU; laptop GPU on Apple Metal; datacenter GPU on NRP/Nautilus · vLLM) produces predictions that flow into a **single evaluation framework whose code is byte-identical for every system**, so any QWK movement is attributable to the model & hardware, not pipeline drift. Results converge on Figure 2 and the results table (Panel 3). Vector diagram, built for print.

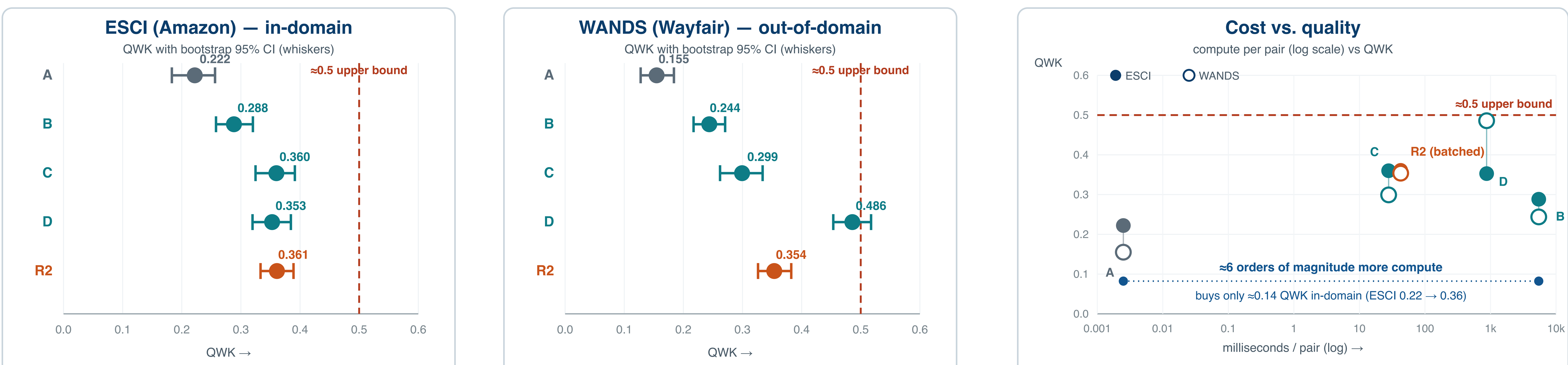

Figure 2. QWK per system with bootstrap 95% CI whiskers on each independent held-out set (**ESCI** in-domain; **WANDS** out-of-domain), and **cost vs. quality** on a log compute axis (filled = ESCI, open = WANDS). The dashed line marks an **empirical upper bound** (≈0.5 QWK) that no system exceeds — not a claim that every system reaches it; a true task ceiling would need human–human agreement (future work). Across **≈6 orders of magnitude** in per-pair compute, in-domain QWK moves only **≈0.14**. R2 is datacenter *batched* throughput, a different latency regime (see L6). Every plotted value is the exact figure from the results table (Panel 3).

0 Abstract

E-commerce search must grade how well each product matches a query on an ordinal **0–3** scale (0 Irrelevant · 1 Marginal · 2 Relevant · 3 Perfect). We ask an honest, reproducible question: **how good a relevance judge can run entirely on a laptop, and when is a GPU actually worth it?**

Using one shared evaluation framework and identical held-out sets (Amazon **ESCI** and Wayfair **WANDS**, 3,000 pairs each, independent human labels), we benchmark six systems across three hardware tiers: a **CPU** lexical baseline (BM25); three **laptop-GPU** systems on Apple Metal (a prompted `llama3.2:3b`, a fine-tuned `bge-reranker-v2-m3` cross-encoder, and a `Llama-3.2-3B` LoRA read via label-token logits); and a **datacenter-GPU** system (`Qwen2.5-7B-Instruct` prompted via vLLM on an NRP/Nautilus GPU).

The headline metric is **Quadratic Weighted Kappa (QWK)** with bootstrap 95% CIs. **Key finding:** every system lands in a narrow **~0.3–0.5 QWK band**. Scaling a prompted judge 3B→7B on a real GPU lifted QWK (+0.073 ESCI, +0.110 WANDS) enough to *match* the fine-tuned cross-encoder in-domain with **no training** — but did **not** break out, and the best out-of-domain result (the 3B LoRA, WANDS 0.486) still stands. **The ceiling is the task, not the model size.** We report every loss beside every win.

1 Background & Problem

- **The task.** A production relevance system (codename *Cortex*) grades each `(query, product title)` pair on the 0–3 `RELEVANCE_SCALE`. Its deployed model is an LLM-as-judge, one model call per pair.
- **The question.** Pointwise relevance grading is an **ordinal classification** problem. How well can it be done on a laptop, and where does reaching for the GPU — first the laptop's integrated GPU, then a datacenter card — actually pay off in accuracy?
- **Why it's hard / matters.** Relevance is **intent, not word overlap**; a "2 vs 1" boundary is genuinely subjective even for humans. An on-device judge is cheaper and private, but must be shown to **beat the local judge it would replace**, on independent labels — not merely reproduce it.

2 Methods

2.1 One shared evaluation framework

Every system runs the identical pipeline so the six are directly comparable. **Inputs:** `(query, product title)` only — the exact signal the production judge uses. **Label mapping** imports `RELEVANCE_SCALE` from the system's own source (never re-typed): ESCI E→3, S→2, C→1, I→0; WANDS Exact→3, Partial→2, Irrelevant→0 (grade 1 deliberately unpopulated).

Metric (headline): QWK — ordinal agreement that penalizes a 3-vs-0 miss far more than a 3-vs-2 miss — with a **1,000-resample bootstrap 95% CI**. Secondary: exact accuracy, ±1 accuracy, MAE, NDCG. Speed and peak RSS measured per system.

2.2 Datasets & avoiding circular evaluation

ESCI (Amazon; native query-disjoint split, verified 0 query overlap) and **WANDS** (Wayfair furniture; eval-only) carry the accuracy verdict — independent human-derived labels. We also keep the system's 4,576 past validations, but use them *only* as a reproduction check, never for accuracy: **92.3%** of those grades were themselves produced by the local `llama3.2:3b`, and here they are graded by a *different* model, so "agreement" there measures whether a model reproduces the local model's pseudo-labels, not whether it is correct. This is a **label-provenance** issue (one model grading another model's labels) — **not** the same as IR "pool bias" or test-collection reusability, where judgments are pooled across systems. Every accuracy claim here is on ESCI/WANDS human labels only.

2.3 The six systems, by hardware tier

Tier	System & approach
CPU	**A — BM25** — classical lexical baseline. CPU, ≈0.003 ms/pair.
Laptop GPU	**B — Prompted `llama3.2:3b`** — LLM-as-judge via Ollama = deployed on-device model. Metal, ≈5,440 ms/pair.
Laptop GPU	**C — bge-reranker-v2-m3** — cross-encoder fine-tuned on ESCI, 4-class argmax (PyTorch-MPS). ≈28 ms/pair.
Laptop GPU	**D — Llama-3.2-3B LoRA** — LoRA fine-tune, read label-token logits (no free generation, Apple MLX). ≈360–873 ms/pair.
Datacenter GPU	**R2 — Prompted `Qwen2.5-7B-Instruct`** — SAME prompt+parser+QWK as System B, bigger model, via vLLM. NVIDIA GPU, batched.

Systems A–D (Round 1) run on an Apple M-series laptop (16 GB); the "GPU" there is the integrated Apple GPU (Metal) — no hand-written CUDA/Metal kernels. **Round-2 (R2)** is the controlled scale-up: it changes **only the model and the hardware**, holding prompt, parser, QWK code and eval sets **byte-for-byte identical** to System B (system-prompt sha256 matched on laptop and in-cluster). Preferred model was **Llama-3.1-8B-Instruct** (clean same-family 3B→8B), but it is gated with no `HF_TOKEN`, so open **Qwen2.5-7B-Instruct** was used (stated confound: family *and* size change). Infra: 1× NVIDIA GPU on NRP/Nautilus via vLLM chat, namespace `kennesaw-state-fjacob`, RTX-3090/A10-class node; model load 103.8 s. Parse failures are excluded from QWK, **never coerced to 0**.

Key finding

Two fine-tunes and a 2.3×-bigger prompted judge all cluster in the **~0.3–0.5 QWK band**. Scaling helped *within* the band but did not break *out* of it — the single best out-of-domain number, the 3B LoRA's **WANDS 0.486**, still stands. **The ceiling is the TASK, not the model size.**

3 Results table — QWK on the independent held-out sets (95% CI)

System	ESCI QWK (95% CI)	WANDS QWK (95% CI)	ESCI acc	WANDS acc
A BM25	0.222 0.183–0.256	0.155 0.128–0.184	0.401	0.377
B Prompted `llama3.2:3b` (deployed)	0.288 0.258–0.320	0.244 0.217–0.271	0.317	0.310
C bge cross-encoder (fine-tuned)	**0.360** 0.325–0.391	0.299 0.262–0.334	0.517	0.473
D Llama-3.2-3B LoRA (logit-read)	0.353 0.320–0.385	**0.486** 0.453–0.517	0.483	0.603
R2 Prompted `Qwen2.5-7B` (GPU)	**0.361** 0.333–0.389	0.354 0.326–0.382	0.343	0.332

`Exact QWK: A 0.22204 / 0.15478 · B 0.28823 / 0.24383 · C 0.35996 / 0.29928 · D 0.35275 / 0.48581 · R2 0.36095 / 0.35367.`

Test sets: 3,000 pairs each. System B parsed 2,993 (ESCI) / 2,989 (WANDS) — 7 / 11 parse failures excluded, never coerced. R2 parsed 2,997 (ESCI) / 3,000 (WANDS) — 3 / 0 failures. Highlighted cells mark the top in-domain (C, ESCI 0.360), the best out-of-domain (D, WANDS 0.486), and the datacenter-GPU in-domain tie (R2, ESCI 0.361). Compare systems WITHIN a dataset only (limitation L2).

4 Results — Round-2 read

Scaling the prompted judge 3B→7B on a datacenter GPU lifted QWK — but stayed in the band. vs System B (same method, 3B): **+0.073 on ESCI** (0.288→0.361) and **+0.110 on WANDS** (0.244→0.354). With **no fine-tuning**, the prompted 7B:

✓ **ties** the fine-tuned cross-encoder on ESCI (0.361 vs 0.360), and
✓ **beats** it on WANDS out-of-domain (0.354 vs 0.299).

But it did NOT break the ceiling. The 7B lands at ~0.35 on both sets — squarely inside the same ~0.3–0.5 band every Round-1 system occupied. A 14B/70B prompted judge was **deliberately not run**: the trend showed diminishing returns inside the band, so more GPU was not justified.

4.3 Speed & memory — why the laptop still wins deployment

System	Latency (ms/pair)	Peak mem
A BM25	≈0.003 lexical, CPU	~0.11 GB
B Prompted 3b	≈5,440 / 5,200 conc-4; ~0.73–0.77 pairs/s	Ollama ~5–6 GB
C bge	**27.9** / 29.6 p95 ~38–40 ms; 0 parse fail	**~1.06 GB**
D LoRA	873 / 360 p95 3,220 / 1,430 ms	~2.67 GB
R2 Qwen 7B	**≈43 / 30** batched; 103.8 s load	≥24 GB GPU

Deployment winner: bge (System C) — matches/beats the on-device judge on accuracy at **~1–2 orders of magnitude faster** (~28 ms vs thousands of ms/pair) in **~5× less memory**, never malforming output. **Out-of-domain winner:** the LoRA (System D) — WANDS 0.486 (a 120-pair re-run reproduced ~0.448); cost ~20–30× slower than bge.

5 Where each system LOST

The project contract: report losses, not just wins.

- A **BM25:** lost everywhere on accuracy — lexical overlap ≠ intent. The floor, kept for reference.
- B **Prompted llama3.2:3b** (deployed model): lost to both fine-tunes on every independent metric and is the **slowest on-device** system. Its only high number — agreement with the system's own past grades — is *self-reproduction*, not accuracy.
- C **bge cross-encoder: LOST out-of-domain** — WANDS 0.299 vs the LoRA's 0.486. Also a train/test gap (dev QWK peaked 0.471 vs held-out ESCI 0.360 — the honest number is the held-out one).
- D **LoRA:** lost on speed/memory (~20–30× slower than bge) and only *tied* bge in-domain on ESCI (0.353 vs 0.360). Its win is generalization.
- R2 **Qwen2.5-7B (GPU):**
  - **Loses to the LoRA out-of-domain: WANDS 0.354 vs 0.486** — that number stands.
  - Loses on **exact accuracy** to both fine-tunes (0.343/0.332 vs bge 0.517/0.473, LoRA 0.483/0.603). Its QWK is respectable only because misses are ordinally *close* (±1 acc 0.79/0.87).
  - **Under-predicts grade 3** (predicts 3 for only 479/2,997 ESCI, 148/3,000 WANDS): agrees on *rank* far better than on *label*.
  - Needs a ≥24 GB datacenter GPU + ~104 s load — bge still wins the efficiency tradeoff.

6 Limitations & honest caveats

- **L1 — Absolute QWK is modest.** Everything sits at ~0.3–0.5: real improvements over baseline, **not a solved problem.**
- **L2 — Compare systems WITHIN a dataset only.** WANDS is a coarser 3-class space (0/2/3, no grade 1), so a WANDS number and an ESCI number are not on the same ruler.
- **L3 — Independent labels = accuracy; system history = reproduction.** 92.3% of past grades came from the same local model — excluded from every accuracy claim.
- **L4 — This is an ON-DEVICE contest.** The system's shipped default is cloud-first (a hosted Gemini model), not benchmarked and likely to beat every laptop-bound system. The claim is "no API, runs on this machine, and beats the local judge" — not "beats frontier cloud models."
- **L5 — Round-2 family confound (open).** Qwen2.5-7B (open) stood in for Llama-3.1-8B (gated), so the +0.073/+0.110 mixes family with size. A clean same-family 3B→8B ablation is still open.
- **L6 — Latency methods differ across systems;** even normalized, the ordering (bge ≪ LoRA ≪ prompted) holds.
- **L7 — 14B/70B not run** — a deliberate choice: the 3B→7B trend already showed diminishing returns inside the band.

7 Conclusion & impact

For an on-device search-relevance judge, **fine-tuning a small specialist (bge cross-encoder) is the best accuracy-per-millisecond** — better than the deployed model, ~1–2 orders of magnitude faster, ~5× lighter.

A bigger *prompted* model on a datacenter GPU buys **in-domain parity with no training**, but not a breakthrough: the **~0.3–0.5 QWK ceiling is a property of the task**, and the best out-of-domain result comes from a **3B LoRA on a laptop**, not from more GPU.

Practical takeaway: reach for the GPU to train a small specialist or to skip training via a mid-size prompted model — but don't expect raw model size to break the quality ceiling on pointwise relevance.

Reproducibility

All systems share one evaluation framework (`scripts/run_arm.mjs`, `src/evaluate.mjs`); label mapping imports the production `RELEVANCE_SCALE`; datasets are public ESCI and WANDS. Round-2 held prompt/parser/QWK byte-identical to Round-1 System B and ran on NRP/Nautilus via vLLM. Journals `00` – `09` document every decision, including the discarded first System-B run and the provider-mix correction. Repo: `experiments/relevancy-replacement/`.